



ASSOCIATION FOR MOLECULAR PATHOLOGY

Providing global expertise in molecular testing that drives patient care
6120 Executive Boulevard, Suite 700, Rockville, Maryland, 20852
Tel: 301-634-7987 | Fax: 301-634-7995 | amp@amp.org | www.amp.org

Department of Health and Human Services
Assistant Secretary for Technology Policy/
Office of the National Coordinator for Health Information Technology (ASTP/ONC)
Mary E. Switzer Building
330 C Street SW, Washington, DC 20201

Electronically submitted via <https://www.regulations.gov/commenton/HHS-ONC-2026-0001-0001>

Re: Docket: HHS-ONC-2026-0001-0001 HHS Health Sector AI Request For Information (RFI):
Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care

To Whom It May Concern:

Thank you for the opportunity to submit comments in response to this Request For Information (RFI): Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care (Docket ID: HHS-ONC-2026-0001-0001). Below is a summary of our comments and requests as they pertain to specific questions in the RFI along with additional background information contained at the end.

The Association for Molecular Pathology (AMP) is an international medical and professional association representing approximately 3,100 physicians, doctoral scientists, and medical laboratory scientists (technologists) who perform or are involved with laboratory testing based on knowledge derived from molecular biology, genetics, and genomics. Membership includes professionals from academic medicine, hospital-based and private clinical laboratories, the government, and the *in vitro* diagnostics industry. AMP members are highly involved in the development, validation, and interpretation of molecular diagnostic tests, including biomarker tests for the diagnosis and management of inherited disorders, cancers, and rare and infectious diseases.

AMP supports the responsible adoption of artificial intelligence (AI) or machine learning tools to improve clinical molecular diagnostics across clinical laboratory workflows spanning analytics, reporting, and interoperability, provided that is consistent with the practice of laboratory medicine including implementation with human-in-the-loop professional oversight, deliberate validation, transparent performance monitoring, and appropriate guardrails to ensure accuracy, equity, patient safety, and regulatory compliance. To ensure these AMP requests that HHS prioritize four actions:

1. Update the Clinical Laboratory Improvement Amendments (CLIA) to revise the definition of “laboratory” to reflect contemporary molecular laboratory practices and add “laboratory analytics” to cover the use of AI tools in clinical molecular diagnostic testing.
2. Provide implementation guidance and program support to adopt structured, standards-based genomic and patient data for interoperability across Electronic Health Record/Laboratory Information System and payers.
3. Address challenges with Prior Authorization (PA) by requiring payers to provide specific, structured denial rationales including coding and policy references and support electronic prior authorization (ePA) to better align processes across Medicare.
4. Promote validated, human-in-the-loop AI by encouraging validation on representative populations, transparent documentation of training and evaluation data and post-deployment monitoring with final clinical responsibility retained by the qualified laboratory professionals and physicians.

AI holds tremendous promise for medicine including in the delivery of molecular laboratory testing. We provide greater details below in response to the specific questions in the RFI.

Q2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care, and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

1. Update CLIA to reflect current laboratory practices

Clinical laboratories play an extremely important role in the U.S. healthcare system, as an astounding 14 billion laboratory tests are conducted annually with 70% of all medical decisions relying on the results of laboratory tests to guide diagnosis or treatments.¹

In order to assist laboratory professionals, a wide range of Artificial Intelligence/Machine Learning (AI/ML) tools are currently used in clinical laboratories. Some technologies, such as algorithms, work in tandem as part of laboratory developed testing procedures while other more complex AI tools are used to obtain patient results to guide treatment options.

Electronic Health Records (EHRs) that are integrated into healthcare systems and utilized by laboratories throughout the country are tools that help streamline certain laboratory operational processes and reduce administrative burdens. Laboratory information systems (LIS) and other laboratory instrument software are critical to laboratory operations because they have many other functions in addition to those in EHRs. The LIS is the core digital infrastructure that manages, tracks, and integrates all operational, analytical, and reporting activities within a clinical laboratory. In molecular diagnostics—including genomic medicine and infectious disease

¹ Centers for Disease Control and Prevention (CDC) <https://www.cdc.gov/lab-week/about-archive.html#:~:text=Today%2C%20medical%20professionals%20continue%20to,diagnosis%20and%20treatment%20of%20diseases>.

testing—the LIS serves as the backbone connecting specimen flow, analytical instruments, interpretation workflows, compliance systems, and EHRs.

AI plays an extremely crucial role molecular pathology because molecular profiles obtained by genomic testing in oncology, medical genetics, and infectious diseases can guide the clinical management of patients in several ways. Genomic information can provide diagnostic, prognostic, and predictive information, identify biomarker-driven targeted therapies, and detect the development of resistance to therapy and antibiotics, among other uses. Large-scale genomic sequencing is also standard of care for the diagnosis and management of many inherited diseases, including developmental delay, neuromuscular disorders, and inherited cardiomyopathies.

It is becoming increasingly common for laboratories to implement Next Generation Sequence (NGS) testing, which analyzes thousands of genes, is becoming increasingly common in laboratories. Clinicians utilize these tests to identify large numbers of genomic alterations in tumors and microbial nucleic acids and their proteins, which are crucial for optimal patient care, and treatment guidelines are developed based on specific molecular findings. For example, the identification of molecular alterations in tumors is now standard of care in non-small cell lung, breast, hematological neoplasms, and colorectal cancers. Therefore, it is imperative to optimize the interpretation and reporting of molecular results among laboratories performing these tests while utilizing various AI tools and working across different AI systems.

Regulatory Oversight of Laboratories through the Clinical Laboratory Improvement Amendment (CLIA) Regulations

Regulatory oversight of clinical laboratories is provided through the Clinical Laboratory Improvement Amendments (CLIA) of 1988 (US Government Publishing Office, Electronic Code of Federal Regulations, (CFR) Title 42, §Part 493.1)²,

In 2023, AMP released an extensive [proposal](#) to clarify the CLIA requirements. Specifically, AMP's proposal includes updating the definition of “laboratory”³ within 42 CFR Part 493 to encompass modern-day laboratory practices that incorporate facilities that perform flow cytometric, molecular, and genomic examinations that may use AI in their analysis. To further reflect present-day laboratory practices, AMP created a definition for “laboratory analytics” that encompasses several different types of AI that are used daily by laboratory professionals.

AMP requests HHS update the current definition of “laboratory” under 42 CFR 493.2⁴ to reflect modern-day laboratory practices and to include a new definition for “laboratory analytics” with the following language :

Section 2.

² http://www.ecfr.gov/cgi-bin/text-idx?SID=1248e3189da5e5f936e55315402bc38b&node=pt42.5.493&rqn=div5#se42.5.493_11

³ [https://www.ecfr.gov/current/title-42/part-493#p-493.2\(Laboratory\)](https://www.ecfr.gov/current/title-42/part-493#p-493.2(Laboratory))

⁴ [https://www.ecfr.gov/current/title-42/part-493#p-493.2\(Laboratory\)](https://www.ecfr.gov/current/title-42/part-493#p-493.2(Laboratory))

Definitions (a) Section 353(a) of the Public Health Service Act is amended to read as follows -- “(a) Definitions (1) As used in this section, the term “laboratory” or “clinical laboratory” means a facility for the biological, microbiological, serological, chemical, immuno-hematological, hematological, biophysical, cytological, pathological, flow cytometric, molecular, genomic, or other examination of either materials derived from the human body or information obtained through analysis of such materials, for screening purposes or for the purpose of providing information for the prognosis, diagnosis, prevention, or treatment of any disease or impairment of, or the assessment of the health of human beings.

(7) As used in this section, the term “laboratory analytics” includes algorithms (including those powered by artificial intelligence or machine learning), software, mathematical formulae, clinical calculations, risk calculations, and other tools as defined by the Secretary that are used as part of a testing procedure.

2. Provide implementation guidance and program support to adopt structured, standards-based genomic and patient data for interoperability across Electronic Health Record/Laboratory Information System and payers.

AMP has acknowledged the critical role of informatics in advancing genomic medicine. The incorporation of genomics into clinical practice is progressing rapidly, with many laboratories using advanced technology for next-generation sequencing (NGS). However, fully utilizing genomic data within the EHR remains challenging because of limitations in interoperability standards and technology. Organizations like AMP are working to develop guidelines to standardize the interoperability of genomic data in EHR systems. Despite the expansion of genomics in medicine, information systems struggle to effectively support genomic workflows from laboratory and patient-facing provider perspectives. The complexity of genomic data, lack of data standards, and insufficient genomic fluency and functionality have resulted in significant gaps between the generation, interoperability, and utilization of genomic data. These gaps pose challenges for laboratory and pathology professionals, clinicians, and patients in generating, communicating, interpreting, and utilizing genomic test results. The AMP EHR Working Group was formed to assess these challenges and propose solutions for the display and use of genomic data in EHRs¹¹. Genomic data can be difficult to locate in EHRs and more difficult to compare across tests and panels over time. Additionally, the AMP recognizes that all components of the software and systems used to generate genomic test results are important.

In 2016, Congress passed the 21st Century Cures Act. While the Act mandated that laboratories and providers facilitate patient access to electronic health information (EHI), it did not include specific, dedicated financial assistance to cover the costs of these technical implementations and could increase financial and administrative burdens on clinical laboratories seeking to

maintain compliance with regulatory requirements.⁵ To achieve data interoperability, laboratories will require funding to ensure that they have the ability to broadly implement and/or replace systems to support interoperability data standards in compliance with regulatory requirements.

Although the use of genomics in medicine is expanding rapidly, information systems are lagging in their ability to support genomic workflows from both the laboratory and patient-facing provider perspectives. The complexity of genomic data, lack of needed data standards, lack of genomic fluency and functionality, and several other factors have contributed to the gaps between genomic data generation, interoperability, and utilization. These gaps pose significant challenges to laboratory and pathology professionals, clinicians, and patients in terms of generating, communicating, consuming, and using genomic test results.

Concerns exist regarding the ability of EHRs to capture complete and accurate patient information, including genomic testing information (Carter et al., 2022, Background Table X, and additional Background information), which may be amplified when utilizing AI systems. For example, because the AI systems used for populating forms for prior authorization pull this information from EHRs that may contain incomplete and/or inaccurate patient and/or genomic testing information, they may erroneously delay or deny appropriate testing. Gaps in data interoperability exacerbate these challenges. Additionally, AI systems must be maintained and updated in real time to ensure that the latest payer policies and practice guidelines are appropriately applied, along with other challenges mentioned in Table X. If AI systems utilize flawed data, rendered downstream decisions may be inappropriate.

3. Address challenges with Prior Authorization (PA)

In 2025, the National Association of Insurance Commissioners conducted a report and found that approximately 75 percent (3 out of 4) health plans use Artificial Intelligence to determine prior authorization approvals.⁶

According to the Kaiser Family Foundation, nearly 50 million prior authorization requests were sent to Medicare Advantage organizations in 2023.⁷

The prior authorization and appeal processes are also slow and cumbersome. A 2024 survey of providers by the American Medical Association demonstrated that providers and their staff spend more than 13 hours/week (nearly two business days) on prior authorizations and

⁵[https://pmc.ncbi.nlm.nih.gov/articles/PMC6134556/#:~:text=The%2021st%20Century%20Cures%20Act%20\(the%20Act\)%2C%20signed%20into,among%20other%20health%2Drelated%20initiatives.](https://pmc.ncbi.nlm.nih.gov/articles/PMC6134556/#:~:text=The%2021st%20Century%20Cures%20Act%20(the%20Act)%2C%20signed%20into,among%20other%20health%2Drelated%20initiatives.)

⁶ <https://content.naic.org/sites/default/files/inline-files/NAIC%20AI%20Health%20Survey%20Report%20.pdf>

⁷ <https://www.kff.org/medicare/issue-brief/nearly-50-million-prior-authorization-requests-were-sent-to-medicare-advantageinsurers-in-2023/#:~:text=The%20vast%20majority%20of%20denied,of%20the%20prior%20authorization%20process.>

complete an average of 41 prior authorizations per week.⁸ Additionally, providers and laboratories often cannot determine the reason for denials.

Policymakers must make several regulatory and statutory changes to ease the burden on providers, streamline the prior authorization and appeals processes, and align Medicare Advantage with Traditional Medicare to ensure coverage and payment for medically necessary care for all Medicare beneficiaries.

AI is being applied to laboratory and/or treating provider-submitted test prior authorizations to automate, accelerate, and optimize the review, approval, and billing-readiness processes. Key cases, how they work, benefits, and risks are summarized in the background information below. Specifically, regarding laboratory testing in molecular medicine and genomics, AI prior authorization faces additional challenges, including complex clinical justification, evolving guidelines, inadequate coding, and payer knowledge gaps (see Background).

AI-driven prior authorization has significant potential to reduce administrative friction and improve test utilization; realizing these benefits requires addressing data quality, interoperability, and regulatory challenges, particularly in specialized areas like genomic medicine. Key considerations to achieve include, but are not limited to, the following:

- Regulatory Compliance: AI-generated content must meet HIPAA, 21st Century Cures Act, and payer-specific requirements.
- Human-In-The-Loop: All AI-drafted narratives should be reviewed and signed by a qualified provider before submission.
- Data quality: Accurate CPT/HCPCS coding, up-to-date payer policies, and complete patient demographics are essential for reliable AI performance.
- Improved EHR Integration: Better structured data capture in EHRs will enhance the ability of AI to extract relevant clinical context.
- Explainable AI: Development of more transparent AI models that can explain denial or appropriateness decisions to clinicians.
- Real-Time Policy Sync: Enables more dynamic integration with payer policies to reduce lag in rule updates.
- Outcome Tracking: AI systems that learn from approved tests and their clinical outcomes to refine appropriateness models over time.

Q6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

4. Promote validated, human-in-the-loop AI

AI is being applied across multiple stages of molecular laboratory workflows, including sample processing and data analysis, result interpretation, and clinical reporting (Background Table X).

⁸ <https://www.ama-assn.org/system/files/prior-auth-reforms-issue-brief.pdf>

These applications aim to improve the accuracy, efficiency, turnaround time, and clinical utility of molecular test results.

Artificial intelligence (AI) has moved from experimental use to routine deployment in several areas of molecular diagnostics, particularly in genomic variant calling, infectious disease sequencing, and laboratory quality monitoring. In specific analytic tasks with well-defined performance metrics and large training datasets, AI has met or exceeded expectations for accuracy and operational efficiency. However, AI has underperformed in areas involving complex biological interpretation (e.g., variant pathogenicity classification, structural variant interpretation, broad antimicrobial resistance prediction, and variant interpretation in the correct clinical context). Multistakeholder engagement is now required to:

1. Promote responsible clinical validation and regulatory clarity.
2. Incentivize high-value AI applications with demonstrable patient impact.
3. Avoid overinvestment in tools that lack generalizability or cost-effectiveness.
4. Ensure interoperability, transparency, and equity in deployment.

AI in molecular diagnostics can deliver when it replaces manual analytic steps, reduces repeat testing, shortens the time to targeted therapy, and/or increases the diagnostic, predictive or therapeutic yield in actionable conditions. Several safeguards are necessary for safe and effective integration into routine clinical practice (see Background). AI underdelivers when biological knowledge gaps limit model reliability, implementation costs exceed operational savings, and/or generalizability across populations is poor. AI in molecular laboratory medicine is not uniformly transformative as its value may be task-dependent (see Background).

Safely and effectively integrating AI into molecular laboratory testing:

- Robust validation: Prospective clinical validation in diverse populations and comprehensive variant sets.
- Transparent methodology: Clear documentation of model training data, performance metrics, and limitations.
- Human-in-the-loop: Pathologists retain final decision-making authority; AI serves as a decision support tool and not a replacement.
- Continuous monitoring: Ongoing performance tracking, error reporting, and model recalibration.
- Ethical oversight: Institutional review, informed consent, and equity assessments for AI deployment.
- Explainable AI (XAI): Development of interpretable models that provide transparent reasoning for variant classifications.
- Standardized data exchange: Implementation of fit-for-purpose genomic data standards to enable seamless AI integration across systems.
- Outcome tracking: Linking AI-driven diagnoses to patient outcomes to validate and refine model performance.

The adoption of AI in molecular diagnostics is rarely driven by a single individual. It typically requires alignment across clinical, laboratory, financial, regulatory, and IT leadership (see Background). Influence varies by organizational structure (academic medical center, community hospital, or integrated delivery network). Strategic investment, regulatory clarity, and outcome-focused validation will determine whether AI in molecular diagnostics reduces costs and improves patient outcomes — or merely adds technological complexity without measurable benefit.

Thank you again for the opportunity to comment on the use of AI in clinical care. AMP is happy to be a resource. If you require additional clinical or policy information to assist you with this request, please contact AMP's Director of Public Policy, Annie Scrimenti, at ascrimenti@amp.org or AMP's Senior Director, Clinical & Scientific Affairs; Director, Clinical Guidance, Dr. Robyn Temple-Smolkin, at rtemple@amp.org.

Sincerely,

Aaron D. Bossler, MD PhD

President, Association for Molecular Pathology

BACKGROUND

Prior Authorization Background

AI in PA use cases:

- Eligibility & coverage determination: Match patient, payer, and policy rules to determine if a test is covered.
- Order appropriateness checking: Compare requested tests with clinical indications, guidelines, and previous results to flag unnecessary or missing tests.
- Prior authorization (PA) form automation: Auto-population of PA forms from EHR and lab systems.
- Clinical justification generation: Drafting evidence-based rationales citing guidelines, prior results, and diagnosis codes.
- Workflow triage and routing: Prioritize urgent requests and route complex cases to specialists or medical directors.
- Predictive denials and cost estimation: Predict the likelihood of denial and estimate patient financial responsibility.
- Decision support for formulary/test selection: Recommend alternative tests or panels with a higher likelihood of approval.
- Audit, reporting, and analytics: Identify denial patterns, provider behavior, and process bottlenecks.

Models and technologies currently used:

- Natural language processing (NLP; named entity recognition and relation extraction) to extract clinical indications, prior results, and text justifications.

- Classification models for predicting approval/denial and required documentation.
- Transformer LLMs for drafting justifications and summarizing records (with guardrails).
- Rules-based engines for explicit payer policies and hard constraints.
- Robotic process automation (RPA) to interact with payer portals where APIs are unavailable.
- Knowledge bases linking guidelines (e.g., clinical society guidance), coverage policies, and prior authorization rules.

Table 1: Challenges Regarding AI Use in Laboratory Testing Prior Authorizations

Challenge	Description
Data Quality Dependencies	AI systems rely on accurate and complete EHR data; poor documentation quality undermines model performance.
Payer Policy Variability	Payer coverage criteria change frequently and vary by plan, keeping AI systems updated is complex.
Black Box Decision Making	Clinicians may not understand why AI flagged a test as inappropriate, thereby reducing trust and buy-in.
Bias in Training Data	Models trained on historical data may perpetuate biases in test ordering or approval patterns.
Interoperability Gaps	Integration between EHRs, lab systems, and payer platforms remains inconsistent, thereby limiting their automation potential.
Privacy & Security	Sharing clinical data with payers and third-party AI vendors raises concerns regarding data security and patient privacy.
Liability & Accountability	Lack of clarity regarding responsibility if AI-driven denials result in delayed care..
Regulatory Oversight	Limited guidance on the validation and monitoring of AI-driven prior authorization systems for safety.

Genomic testing-specific challenges:

- **Complex Clinical Justification:** Genomic tests often require nuanced clinical reasoning that may be difficult for AI to capture from EHR data alone.
- **Evolving Guidelines:** Genomic testing recommendations change rapidly as evidence accumulates, making it difficult for AI models to stay current.
- **Payer Knowledge Gaps:** Many payers lack expertise in genomics, which may lead to conservative approval criteria that AI systems may apply too rigidly.

Fit-for-purpose Data Interoperability Background

The use of genomics in clinical medicine is expanding rapidly, and many laboratories are implementing highly complex technologies to perform next-generation sequencing (NGS). However, the full utilization of complex genomic data in the electronic health record (EHR) is challenging in the current state of interoperability standards and technology. AMP and other professional organizations are actively working to develop guidance to standardize the interoperability of genomic data in EHR systems.

Genomic data can also be difficult to locate in electronic health records (EHRs) and can be even more difficult to compare across different tests and panels over time. Additionally, AMP is cognizant that all components of the software and systems used to generate genomic test results for patients are important. AMP has long recognized the importance of informatics to the success of genomic medicine.

With the increasing use of molecular pathology and genomic testing, complaints regarding the significant lack of functionality that necessitates the use of paper records and/or manual transfer of data to and from instrument software, laboratory information systems (LISs), and electronic health records (EHRs) have increased. LISs and other laboratory instrument software are critical to laboratory operations because they have many other functions in addition to those in EHRs. The LIS is the core digital infrastructure that manages, tracks, and integrates all operational, analytical, and reporting activities within a clinical laboratory. In molecular diagnostics—including genomic medicine and infectious disease testing—the LIS serves as the backbone connecting specimen flow, analytical instruments, interpretation workflows, compliance systems, and EHRs. Reporting requirements for different testing methods (molecular pathology, genomic and germline and somatic variant) in various types of laboratories were examined with regard to challenges in EHRs, including the following laboratories: molecular and genetic; cytogenetics; those performing fluorescence *in situ* hybridization (FISH), tissue typing, histocompatibility, immunogenetics (human leukocyte antigen (HLA) testing), and infectious organisms.

Utilizing the EHR Working Group’s subject matter expert analysis¹¹, several concerns regarding the use of AI can be extrapolated:

TABLE 2: How EHR Challenges Impact Concerns Regarding Artificial Intelligence Use in Clinical Care for Genomic Medicine

EHR CHALLENGES <i>(Reproduced from Carter et al, 2022)</i>	Concerns Regarding Artificial Intelligence Use
EHRs are not yet ready to send accurate, coded and appropriately granular clinical history, signs, symptoms, family history and other broad sets of data elements to laboratories without generating significant burden on providers.*	LLMs trained or prompted with incomplete/noisy/biased inputs produce incorrect inferences, poor context for variant interpretation, and unreliable recommendations
EHRs lack sufficient information about genetic test orders and generally do not have discrete variant result data to facilitate appropriate test ordering and utilization.*	LLMs cannot map tests to appropriate interpretation logic or link variants to prior results; automated workflows break

Absence of standardized genomic variant data structures from the LIS to the EHR.*	
Interoperability standards for genomics are currently limited in several ways including the use of syntax with limited hierarchy (HL7 version 2.x) and inadequate coding systems for genomic orders and results.*	LLM-driven automation relying on standards will fail across sites; model generalization is reduced and needs complex normalization layers
Genomic reports between laboratories are variable in structure and content.* Implementation of molecular pathology and genomics professional societies multi-organizational consensus standards and guidelines for variant nomenclature, hierarchical result structure, and genomic report formats has been modest.*	LLMs see multiple naming conventions and result hierarchies, harming mapping to canonical knowledge and CDS
There are no standards for how to display aggregated variant data over time and between sample sources, tests and laboratories which keeps clinical context and associated interpretation intact.*	LLMs can't reliably synthesize longitudinal summaries or explain provenance; temporal reasoning degrades
Consensus guidelines on best practice for requesting and providing reclassification of variants is not currently available.*	LLMs cannot automate or advise on reclassification workflows or notifications safely
Genomic reports are difficult for medical personnel and patients to understand, and in the US, patients have the right to immediately access their genomic test reports upon request.*	LLMs may simplify or summarize incorrectly; patient-facing outputs risk oversimplification or alarming language
Absence of national and international standards for Clinical Decision Support (CDS) rules.	LLMs powering or generating CDS lack interoperable rule representations and may produce non-actionable or conflicting guidance
Currently, textual data from genomic reports is exported and released to authorized parties, and analysis of such data is limited by its highly variable format and lack of structure.	LLMs must rely on brittle NLP extraction pipelines, increasing hallucinations and missed structured facts
Current technology lacks sufficient functionality to ensure that release of genomic data to appropriate healthcare organizations, research studies, or clinical repositories requires and receives informed consent from the patient or legal guardian, and/or institutional board review.	LLM-driven data sharing or model training risks using data without proper consent or tracking provenance
Current coding and interoperability standards are not adequate for genomic data.*	LLMs cannot depend on consistent identifiers or mappings to reference databases, reducing accuracy of variant interpretation

LEGEND: This table has been modified from *Electronic Health Records and Genomics: Perspectives from the Association for Molecular Pathology Electronic Health Record (EHR) Interoperability for Clinical Genomics Data Working Group* (Table 1: A Summary of Challenges and Opportunities)¹³. *Areas where the Association for Molecular Pathology is actively engaged with subject matter expert working groups addressing identified challenges and opportunities. Future recommendations, along with accompanying educational offerings, are being developed by the EHR Working Group and other AMP working groups to address these community needs. Abbreviations: EHR - Electronic Health Record; FHIR - Fast Healthcare Interoperable Resources; LIS - Laboratory Information System; LLM – Large Language Models.

Human-In-the-Loop Background Information for #6:Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

Laboratory directors and clinical teams working with medical team senior leadership that have identified ways for AI to help reduce the administrative burden and/or improve patient care. Most influential decision-makers:

1. Laboratory Medical Director (analytic authority)
2. Chief Medical Officer or CMO (clinical governance)
3. Chief Financial Officer or CFO (financial gatekeeper)
4. Chief Information Officer/Chief Data Officer or CIO/CDO (technical feasibility)
5. Stewardship or genetics committees (domain endorsement)
6. External regulators (structural influence)

In most large academic centers, the Laboratory Medical Director + CFO + CIO + CMO alignment is the critical adoption bottleneck. If the laboratory director is convinced of analytic validity and the CFO sees cost justification with CMO endorsement required for standard-of-care change, adoption is likely. Without alignment among these four, deployment rarely progresses beyond limited pilots. Primary administrative hurdles include, but are not limited to, implementation, data storage, and maintenance costs in both direct costs and FTE; ongoing personnel and direct costs to support validation and use of AI in a CLIA environment; and additional education to understand the complexity of the test systems and inputs needed to build, launch, and operate AI in a manner that reduces the risk of patient harms and medical errors. Other hurdles include regulatory ambiguity, reimbursement gaps, integration complexity, validation burden, workforce trust, infrastructure cost, and fragmented governance.

AI in Molecular Laboratory Medicine Emerging High-Value Areas

- o Deep learning variant calling.
- o AI-augmented pathogen detection.
- o Rapid genomic AST in defined organisms.
- o Laboratory process monitoring.
- o Supervised automated laboratory document (SOP, validation summary) generation using LLM

AI in Molecular Laboratory Medicine Areas Requiring Caution

- o Stand-alone prediction of sequence variant pathogenicity.
- o Broad-spectrum antimicrobial resistance modeling.
- o Interpretation of structural or non-coding variants without functional validation.
- o Interpretation of genomic alterations in complex neoplastic and non-neoplastic clinical contexts
- o Prediction of therapy effectiveness based on genomic variants

Table 3: AI Applications Across the Molecular Laboratory Workflow

Stage of Workflow	AI Application	Description	Benefit
Sample Processing & QC	Automated image analysis	Computer vision algorithms analyze gel electrophoresis images, sequencing runs, and PCR products for quality assessment.	Reduces manual review time, improves consistency, and automatically flags quality issues.
Sample Processing & QC	Robotic process automation	AI-guided robotic systems can handle sample preparation, DNA/RNA extraction, and library preparation with minimal human intervention.	improves throughput, reduces contamination risk, and improves reproducibility.
Sequencing Data Analysis	Variant calling optimization	Machine learning models improve the accuracy of variant detection from raw sequencing data (NGS).	Reduces false positives/negatives and improves sensitivity for low-frequency variants.
Sequencing Data Analysis	Copy number variation (CNV) detection	AI algorithms identify deletions, duplications, and structural variants based on the sequencing depth and read alignment patterns.	Detects variants missed by traditional methods and improves diagnostic yield.
Data Analysis	Quality filtering & normalization	AI models learn patterns in sequencing noise and systematically remove artifacts while preserving true variants.	Improves data quality and reduces the manual curation burden.
Variant Interpretation	Pathogenicity prediction	Machine learning models (e.g., SIFT, PolyPhen, CADD, AlphaMissense) predict whether variants are likely pathogenic based on sequence context and evolutionary conservation.	Aids interpretation of variants of uncertain significance (VUS), accelerates reporting.

Variant Interpretation	Functional prediction	Deep learning models predict protein structure changes and functional consequences of missense variants.	Improves variant classification, supports clinical decision-making.
Variant Interpretation	Literature & database mining	NLP algorithms extract variant–disease associations from scientific literature and integrate curated databases (ClinVar, OMIM, gnomAD).	Provides comprehensive evidence for variant classification and reduces manual literature review requirements.
Clinical Interpretation	Phenotype-genotype matching	AI systems match patient phenotypes to genetic variants and known disease associations.	Identifies causative variants and suggests differential diagnoses, thereby improving diagnostic accuracy.
Clinical Interpretation	Gene-disease association discovery	Machine learning identifies novel or emerging gene–disease relationships from large datasets.	Expands knowledge of disease genetics and supports research and discovery.
Report Generation	Automated report writing	NLP and template-based systems generate structured clinical reports with interpretations and recommendations.	Standardizes reports, reduces turnaround time, and ensures consistent formatting.
Report Generation	Clinical decision support (CDS)	AI systems provide actionable recommendations based on variant interpretation, patient context, and clinical guidelines.	Guides clinicians on next steps (e.g., confirmatory testing, family screening, and therapy options).
Post-Analysis	Variant reclassification monitoring	AI tracks published evidence and flags variants for reclassification when new data emerges.	Ensures patients receive updates on variant classification, improves long-term care.
Post-Analysis	Incidental findings detection	Machine learning identifies secondary findings or variants with clinical significance outside the primary indication.	Supports the return of incidental findings and improves preventive care opportunities.

There are several potential benefits of AI in clinical molecular laboratories

- **Improved Diagnostic Accuracy:** AI enhances variant detection and interpretation, thereby reducing missed diagnoses and false positives.
- **Faster Turnaround Times:** Automation and intelligent prioritization reduce the time from sample to report.
- **Increased Throughput:** Robotic and automated workflows allow laboratories to process more samples efficiently.
- **Improved Workflows:** Automation decreases manual curation and review work.
- **More Standardized Reporting:** AI-generated reports ensure consistency and compliance with professional guidelines.

- Knowledge Integration: NLP and machine learning synthesize vast amounts of genetic and clinical literature automatically.
- Proactive Variant Monitoring: AI tracks evolving evidence and alerts clinicians to reclassifications or new findings.

Table 4: Challenges and Concerns Specific to Molecular Labs

Challenge	Description	Impact
Model validation & generalization	AI models trained on one population may not perform well on diverse genetic backgrounds or rare variants.	Reduced accuracy in underrepresented populations; potential diagnostic disparities.
Insufficient training data	Rare variants and disease associations have limited training examples, making model predictions unreliable.	VUS remain difficult to classify; models may overfit to limited data.
Lack of standardized genomic data structures	Inconsistent variant nomenclature and data formats hamper AI integration.	Models must handle variable inputs; interoperability between laboratories and systems is poor.
Explainability & clinical acceptance	Clinicians need to understand why AI flagged a variant as pathogenic; black-box models reduce trust.	Pathologists may override or distrust AI recommendations; liability concerns arise.
Regulatory uncertainty	FDA guidance on AI/ML in clinical genomics is evolving; validation and oversight standards remain unclear.	Laboratories uncertain about compliance requirements; slow adoption of validated AI tools.
Data privacy & consent	Training AI on genomic data raises privacy concerns; informed consent requirements are uncertain.	Legal/ethical risks; patient hesitation; regulatory scrutiny.
Variant interpretation complexity	Many variants lack functional data; AI predictions may not capture biological complexity.	VUS remain challenging; over-reliance on prediction scores can mislead clinicians.
Integration with EHR/LIS	Genomic AI outputs must integrate with LIS and EHR systems, but interoperability is limited.	Manual data transfer, fragmented workflows, increased error risk.
Bias in training datasets	Historical classifications and datasets may embed population biases.	Disparities in diagnostic accuracy, especially for non-European ancestry groups.
Cost of implementation	Developing, validating, and maintaining AI systems is resource-intensive.	Widening gap between well-resourced vs. under-resourced laboratories; consolidation pressures.
Keeping models current	Rapidly evolving genomic knowledge requires continuous updates and revalidation.	Outdated interpretations; high ongoing maintenance burden.

For AI to be safely and effectively integrated into molecular laboratory testing:

- Robust validation: Prospective clinical validation on diverse populations and comprehensive variant sets.
- Transparent methodology: Clear documentation of model training data, performance metrics, and limitations.
- Human-in-the-loop: Pathologists retain final decision-making authority; AI serves as decision support, not replacement.
- Continuous monitoring: Ongoing performance tracking, error reporting, and model recalibration.
- Ethical oversight: Institutional review, informed consent, and equity assessments for AI deployment.
- Explainable AI (XAI): Development of interpretable models that provide transparent reasoning for variant classifications.
- Standardized data exchange: Implementation of fit for purpose genomic data standards to enable seamless AI integration across systems.
- Outcome tracking: Linking AI-driven diagnoses to patient outcomes to validate and refine model performance.

AI holds tremendous promise for molecular laboratory testing, but realizing its benefits requires addressing data standardization, validation rigor, and data interoperability challenges—particularly for rare variants and underrepresented populations.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Adoption of AI in molecular diagnostics is rarely driven by a single individual. It typically requires alignment across clinical, laboratory, financial, regulatory, and IT leadership. However, influence varies by organizational structure (academic medical center vs. community hospital vs. integrated delivery network).

Laboratory directors and clinical teams working with medical team senior leadership that have identified ways for AI to help reduce administrative burden and/or improve patient care. Most influential decision makers:

7. Laboratory Medical Director (analytic authority)
8. Chief Medical Officer (CMO; clinical governance)
9. Chief Financial Officer (CFO; financial gatekeeper)
10. Chief Information Officer/Chief Data Officer (CIO/CDO; technical feasibility)
11. Stewardship or genetics committees (domain endorsement)
12. External regulators (structural influence)

In most large academic centers, the Laboratory Medical Director + CFO + CIO + CMO alignment is the critical adoption bottleneck. If the laboratory director is convinced of analytic validity and the CFO sees cost justification with CMO endorsement required for standard-of-care change, adoption is likely. Without alignment among these four, deployment rarely

progresses beyond limited pilots. Primary administrative hurdles include but are not limited to implementation, data storage, and maintenance costs in both direct costs and personnel full time equivalents (FTE); ongoing personnel and direct costs to support validation and use of AI in a CLIA environment; additional education to understand the complexity of what systems and inputs will be needed to build, launch, and operate AI in a manner that reduces the risk of patient harms and medical errors. Other hurdles include regulatory ambiguity, reimbursement gaps, integration complexity, validation burden, workforce trust, infrastructure cost, and fragmented governance.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

More robust information provided to cancer registries and other publicly available HIPAA-compliant / deidentified databases will fuel ongoing research and biomarker identification using AI pattern recognition; real-world evidence from EHRs will be less prone to data dropouts and potential errors;

AI in genomic medicine and infectious disease remains less constrained by algorithmic sophistication and more by data fragmentation, inconsistent standards, and a lack of cross-institutional benchmarking infrastructure.

Enhanced interoperability would:

- Unlock larger, higher-quality training datasets
- Improve generalizability and regulatory readiness
- Reduce vendor lock-in
- Enable multi-site benchmarking
- Expand reimbursement viability
- Widen total addressable market for AI vendors
- Expand market opportunity
- Reduces development cost
- Increase model robustness
- Improve patient outcomes
- Enhance public health response

Table 5: Examples of Interoperability Opportunities in Genomic Medicine & Infectious Disease

Category	Sub-Area	Current Limitation / Gap	High-Impact Data Types	Key Standards	AI / Market Impact

Genomic Medicine	Variant-Level Genomic Data Standardization	Variant data stored in VCF, LIS, PDFs, and proprietary systems; not machine-readable; inconsistent across institutions.	VCF files; BAM/CRAM; CNV data; structural variant calls; variant metadata; zygosity; read depth; quality metrics.	HL7 FHIR Genomics; GA4GH VRS; GA4GH Beacon; ClinVar standards; HGVS.	Enables cross-institution AI training; reduces validation burden; enables federated learning; real-time reclassification alerts; supports regulatory benchmarking; accelerates pathogenicity/structural variant models and rare disease discovery.
Genomic Medicine	Genotype–Phenotype Integration	Phenotype data stuck in EHR notes, ICD-10 codes, lab tables, imaging systems; poor interoperability.	HPO terms; SNOMED CT diagnoses; longitudinal laboratories; medications; family history; social determinants.	FHIR Clinical Genomics; HPO; OMOP; SNOMED CT; LOINC.	Enables genotype–phenotype AI; improves rare disease algorithms; enhances penetrance/risk prediction; supports precision diagnostics and CDS; and expands datasets for vendors.
Genomic Medicine	Structural Variant & Multi-Omics Interoperability	SVs and multi-omics (RNA-seq, methylation, proteomics and metabolomics) are siloed; inconsistent data formats.	RNA-seq; methylation profile; proteomic; metabolomic; and structural variant representations.	GA4GH Phenopackets; standardized SV formats; FAIR principles.	Improves diagnostic yield for unsolved cases; supports cancer predisposition modeling; enhances functional annotation AI.
Infectious Disease	AMR Genotype–Phenotype Databases	Genomic and resistance phenotype data are disconnected and have limited linkage to outcomes.	MICs, antibiotic exposure, resistance genes, plasmid detection, patient outcomes, and treatment timing.	CLSI; EUCAST; WHO GLASS; FHIR MedicationStatement; standardized AST ontologies.	Improves genotype-to-phenotype prediction; supports generalizable organism-specific AI; enables global surveillance; expands ROI and reimbursement potential.
Infectious Disease	Real-Time Infection Surveillance Networks	Infection data siloed in microbiology laboratories, infection prevention, and state registries.	Time-stamped cultures, pathogen WGS, outbreak clusters, patient movement, and environmental samples.	FHIR Public Health, HL7 ELR, and PulseNet.	Enables real-time outbreak detection, predictive transmission modeling, and early AMR warning systems.

Abbreviations: VCF — Variant Call Format, LIS — Laboratory Information System, PDF — Portable Document Format, BAM — Binary Alignment/Map, CRAM — Compressed Reference-oriented Alignment Map (format), CNV — Copy Number Variant, HL7 — Health Level Seven International, FHIR — Fast Healthcare Interoperability Resources, GA4GH VRS — Global Alliance for Genomics and Health, Variation Representation Specification, HGVS — Human Genome Variation Society, EHR — Electronic Health Record, HPO — Human Phenotype Ontology, ICD-10 — International Classification of Diseases, 10th Revision, OMOP — Observational Medical Outcomes Partnership (Common Data Model), SNOMED CT — Systematized Nomenclature of Medicine — Clinical Terms, LOINC — Logical Observation Identifiers Names and Codes, AMR — Antimicrobial Resistance, CLSI — Clinical and Laboratory Standards Institute, EUCAST — European Committee on Antimicrobial Susceptibility Testing, WHO GLASS — World Health Organization Global Antimicrobial Resistance and Use Surveillance System, FAIR Principles — Findable, Accessible, Interoperable, and Reusable